Regional Studies, Regional Science

ISSN: 2168-1376 (Online) Journal homepage: www.tandfonline.com/journals/rsrs20

Scotland's twin referendums: revealing regional divides behind national votes

Craig Stewart, Xinyi Yuan & Mingshu Wang

To cite this article: Craig Stewart, Xinyi Yuan & Mingshu Wang (2026) Scotland's twin referendums: revealing regional divides behind national votes, Regional Studies, Regional Science, 13:1, 2637381, DOI: 10.1080/21681376.2026.2637381

To link to this article: https://doi.org/10.1080/21681376.2026.2637381

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

Published online: 16 Mar 2026.

Submit your article to this journal

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=rsrs20

REGIONAL STUDIES, REGIONAL SCIENCE
2026, VOL. 13, NO. 1, 2637381
https://doi.org/10.1080/21681376.2026.2637381

Regional Studies Association
Routledge Taylor & Francis Group

REGIONAL GRAPHIC

OPEN ACCESS Check for updates

Scotland's twin referendums: revealing regional divides behind national votes

Craig Stewart, Xinyi Yuan and Mingshu Wang

School of Geographical and Earth Sciences, University of Glasgow, Glasgow, UK

ABSTRACT
Scotland's 2014 independence referendum and the 2016 EU referendum are often interpreted as politically aligned at the national scale, reinforcing a simplified narrative of a strongly pro-independence, pro-EU portion of the Scottish electorate. This regional graphic challenges that view using a bivariate, population-weighted cartogram that jointly visualises both votes across parliamentary constituencies. By correcting for uneven population distributions and revealing sub-regional variation, the graphic exposes multi-scalar political cleavages within cities, regions and economic hinterlands. The visualisation demonstrates how national aggregates obscure regional political heterogeneity and highlights the value of spatially explicit graphics for interpreting constitutional and electoral change.

ARTICLE HISTORY
Received 29 December 2025
Accepted 23 February 2026

KEYWORDS
Electoral geography; regional political geography; population-weighted cartogram; Scotland

1. Main text

Scotland's 2014 independence referendum delivered a 55.3% vote to remain in the UK (McInnes et al., 2014). Two years later, the UK-wide Brexit referendum returned a narrow 52% majority for leaving the EU (Uberoi, 2016), while Scotland voted 62% Remain. These back-to-back events exposed clear geographic and political contrasts within the UK and within Scotland itself. Yet public debate quickly compressed these differences into simple narratives: such as pro-independence voters also being strongly pro-EU. The spatial reality is more fragmented.

Since 2016, independence and EU membership have often been portrayed as aligned expressions of a liberal, outward-looking Scotland. While this alignment became politically salient after Brexit, it did not fully exist beforehand. British Election Study data show that 38% of 2014 Yes voters supported Leave in 2016 (Curtice, 2020). Only later did pro-EU sentiment consolidate among independence supporters, reaching around 59% by 2023 (Curtice, 2024). Political identities, therefore, evolved in response to Brexit rather than preceding it.

Local variations complicate the narrative further. National aggregates obscure the intra-urban and intra-rural heterogeneity (Neal et al., 2021). Moray, historically tied to fishing, voted close to the national average in the 2014 independence referendum but showed markedly higher support for Brexit in 2016, reflecting longstanding discontent with the EU's Common Fisheries Policy (Agnisola et al., 2019). In contrast, parts of Glasgow combined strong independence sentiment with pockets of Leave support in post-industrial neighbourhoods. Edinburgh displayed low support for both Yes and Leave, reflecting a different economic and demographic profile. These examples disrupt the simplistic 'urban = Yes/Remain, rural = No/Leave' division.

To make these nuances visible, we present a bivariate population-weighted cartogram comparing both referendums across Scotland (Figure 1). Each unit is resized in proportion to its 2011 Census population, addressing distortions created by uneven constituency boundaries. Using established downscaling approaches (Hanretty, 2017; Miori, 2024), we re-estimated referendum outcomes at the parliamentary constituency level, producing more population-balanced spatial units and revealing finer-grained structure. At a broader scale, NUTS-2 (nomenclature of territorial units for statistics – level 2) regions highlight how

CONTACT Mingshu Wang mingshu.wang@glasgow.ac.uk

© 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group
This is an Open Access article distributed under the terms of the Creative Commons Attribution License (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

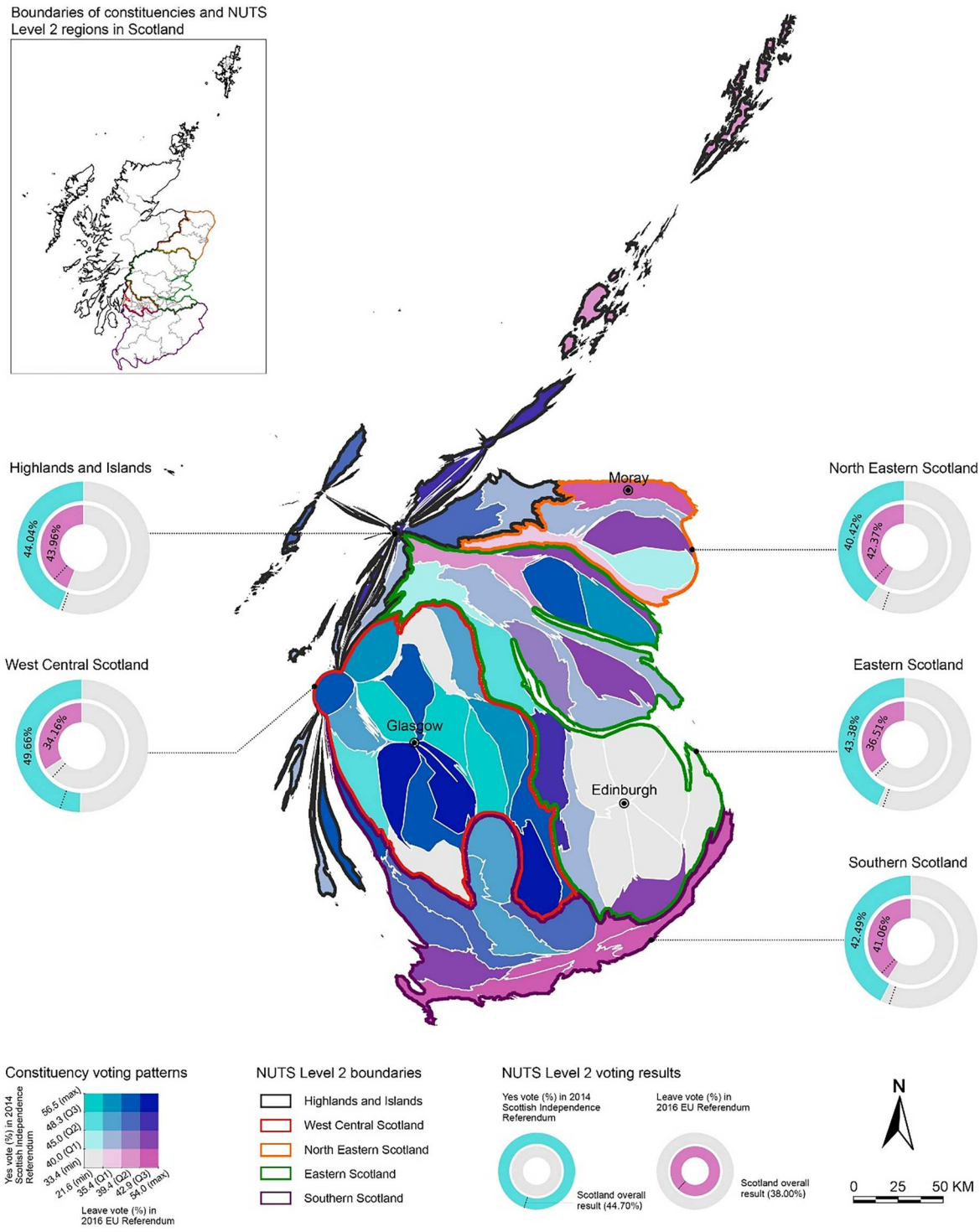

Figure 1. Population-weighted spatial relationship between Scotland's 2014 independence referendum and the 2016 EU referendum.

areas beyond the central belt, particularly the Highlands and Islands and North East Scotland, leaned more strongly toward Leave, likely reflecting agricultural and fisheries-based economic dependencies.

The resulting graphic shows that spatial cleavages in Scottish political behaviour are multi-scalar, operating simultaneously across neighbourhood, city and regional levels. Binary referendums compress multi-dimensional identities, masking how economic structure, local history and regional context shape political behaviour. Scotland's twin referendums illustrate why regional and spatially explicit visualisation is essential for interpreting political change and anticipating future constitutional debates.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the Alan Turing Institute (grant number 2024 Enrichment Scheme Community Award).

Data availability statement

All data used in this study are derived from publicly available sources. Official referendum results for the 2014 Scottish Independence Referendum and the 2016 EU Membership Referendum were obtained from the UK Electoral Commission.

ORCID

Craig Stewart https://orcid.org/0009-0006-9473-2043
Xinyi Yuan https://orcid.org/0009-0003-7790-3662
Mingshu Wang http://orcid.org/0000-0001-5260-3143

References

Agnisola, G., Weir, S., & Johnson, K. (2019). The voices that matter: A narrative approach to understanding Scottish Fishers' perspectives of Brexit. *Marine Policy*, *110*, 103563. https://doi.org/10.1016/j.marpol.2019.103563

Curtice, J. (2020). High noon for the Union? *IPPR Progressive Review*, *27*(3), 223–234. https://doi.org/10.1111/newe.12217

Curtice, J. (2024). *How Brexit has changed Scotland's constitutional debate*. Scottish Centre for Social Research.

Hanretty, C. (2017). Areal interpolation and the UK's referendum on EU membership. *Journal of Elections, Public Opinion and Parties*, *27*(4), 466–483. https://doi.org/10.1080/17457289.2017.1287081

McInnes, R., Hawkins, O., & Ayres, S. (2014). Scottish independence referendum 2014 analysis of results. *House of Commons Library*. https://commonslibrary.parliament.uk/research-briefings/rp14-50/

Miori, M. (2024). Notional 2014 Scottish independence referendum results for 2024 UK Constituencies. *University of Manchester*. Dataset. https://doi.org/10.48420/26340568.v1

Neal, S., Gawlewicz, A., Heley, J., & Jones, R. D. (2021). Rural Brexit? The ambivalent politics of rural community, migration and dependency. *Journal of Rural Studies*, *82*, 176–183. https://doi.org/10.1016/j.jrurstud.2021.01.017

Uberoi, E. (2016). EU referendum: Results and analysis. House of Commons Library Briefing Paper CBP-7639. http://researchbriefings.files.parliament.uk/documents/CBP-7639/CBP-7639.pdf